MILITARY SPECIFICATION

CHEST: COLLAPSIBLE TANK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a skid mounted aluminum alloy chest.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/3	- Aluminum Alloy Bar, Rod, Shapes, and Tubes, Extruded 2024.
QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes, and Tube, Extruded, 6061.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.

FSC 5430

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-W-470	- Wire, Steel, Carbon, Spring, Music.
QQ-Z-325	- Zinc Coating, Electrodeposited, Requirements for.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-1149	- Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-15024/6	- Plate, Identification.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED. TEST METHOD No. 101	- Preservation, Packaging, and Packing Materials; Test Procedures.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

DRAWINGS

ME

TA13222E7930	- Chest, Collapsible Tank.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Description. The chest shall be in accordance with Top Assembly TA13222E7930 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the chest. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced chest). The contractor shall furnish one or more chests as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce chests that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.3.1 Plate and sheet aluminum alloy. Plate and sheet aluminum alloy for the chest sides, ends, lid and bottom shall conform to QQ-A-250/11, Class T6.

3.3.2 Aluminum bars, rods and shapes. Aluminum bars, rods and shapes, extrusions for the lid packing gland, stiffeners, corner posts and skids shall conform to QQ-A-200/8, Class T6, or QQ-A-200/3, Class T6. The extrusion shall have rounded edges with material thickness not less than 0.090 inch. The carrying bars shall be not less than 0.75 inch diameter.

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3.3.3 Springs. Springs shall be steel wire conforming to QQ-W-470 and shall be zinc plated in accordance with QQ-Z-325, Type I.

3.3.4 Gasket material. Gasket material shall conform to MIL-G-1149, Type I, Class 5.

3.4 Screw threads. All screw threads shall be in accordance with FED-STD-H28.

3.5 Construction.

3.5.1 Weight. The empty chest, complete with all components, shall weigh not more than 325 pounds.

3.5.2 Lid fastening devices. Seven latches shall be provided on one side of the chest. The latches shall be spring loaded, quick acting, dog type, and shall be self positioning along the chest side when not in use. Seven hold-down devices, attached to the side of the chest opposite the latches, shall provide a positive hold for the lid in the longitudinal, transverse, and vertical directions. The hold-down devices shall permit rapid removal of the lid, and shall function when the lid is installed in any given mating position.

3.5.3 Storage compartment. A 10-inch-wide compartment, for storing tank fittings and the tank repair kit, shall be provided at one end of the chest.

3.5.4 Skids. Two skids, not less than 2.5-inches wide shall be provided on the bottom of the chest. The skids shall run the entire length of the chest and shall be turned up at each end not less than 3 inches on a 14- to 16-inch radius. The skid webs shall be slotted to permit handling the chest by forklift truck.

3.5.5 Stacking lugs. Four stacking lugs shall be provided on the top of the chest lid for use as a stacking aid.

3.6 Performance.

3.6.1 Lid operation. The lid shall be installed or removed from the chest manually by not more than two men without the use of tools. The lid when installed and latched in place shall provide a seal top to protect the chest contents during storage, handling, and shipping.

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3.6.1.1 End-for-end rotation. The lid lugs shall engage the lid fastening devices as specified in 3.5.2 when turned end-for-end (180 degrees rotation in the horizontal plane).

3.6.2 Proof load. There shall be no damage or permanent distortion to the chest, lid, lifting hooks, or failure of the latches when loaded with a gross weight of 2,600 plus or minus 10 pounds, lifted free of all other support by the four lifting hooks, and suspended for a period of not less than 10 minutes.

3.6.3 Carrying bars. The chest, loaded with a gross weight of 1,200 plus or minus 5 pounds, and the lid latched in place shall withstand an upward pull of 300 plus or minus 3 pounds, on either carrying bar, without damage or permanent distortion to the chest, carrying bar, lid, or failure of the latches.

3.6.4 Drop. The chest, loaded as specified in 3.6.3 shall show no failure of the latches, nor damage or permanent distortion to the chest, or lid, when tested as specified in 4.5.2.4. Repairable, superficial damage will be permitted.

3.6.5 Edgewise drop. The chest, loaded as specified in 3.6.3, shall show no failure of the latches nor damage or permanent distortion to the chest, lid, or skids, when tested as specified in 4.5.2.5. Repairable, superficial damage will be permitted.

3.6.6 Towing. The chest, loaded as specified in 3.6.3, towed from either end by use of the towing eyes for a distance of 400 plus or minus 5 feet across concrete pavement at a speed of 5 to 7 miles per hour, shall show no failure of the latches nor damage or permanent distortion to the chest, lid, or towing eyes. Repairable, superficial damage will be permitted.

3.7 Identification marking. The chest shall be identified in accordance with MIL-STD-130. The marking shall be applied to the chest on plates conforming to MIL-P-15024/6. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.8 Treatment and painting. The portions of the chest normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color forest green. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

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3.8.1 Anodic coating. All aluminum surfaces not painted shall have rough and sharp edges and scale removed. These surfaces shall be cleaned, anodized, and dyed in accordance with MIL-A-8625, Type II, Class 2. The color shall be dark, nonvivid, and nonreflective.

3.9 Fungus-inert material. The chest shall be constructed and treated to provide protection for the chest contents against fungus and moisture damage.

3.10 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts and without selection for fit or performance.

3.11 Workmanship. Workmanship shall not permit defects adversely affecting the strength or serviceability of the chest. All metal parts shall be cleaned and free of sand, dirt, scale, and flux. Surfaces shall be smooth with edges rounded or beveled. The inside and outside of the chest shall be clean and free of foreign materials. Any necessary repair or rework shall restore the reworked area to its full strength and shall meet all applicable requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.1.2 Disassembly inspection. Failure of any examination or test by the first produced chest shall be cause for disassembly, in the presence of a Government representative, of the first produced chest to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced chest. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced chest inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced chest inspection.

4.3.1 Examination. The first produced chest shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first produced chest shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for defects specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

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4.5.1 Examination. The chest shall be examined as specified herein for the following defects:

101. Any part (or component) not as shown on the drawings.
102. Dimensions not as specified.
103. Materials not as specified.
104. Weight not as specified.
105. Components missing or not as specified.
106. Color not as specified.
107. Identification marking missing, incomplete, or illegible.
108. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Lid operation. Lid operation shall be tested as follows:

(a) Install lid and latch into place. Visually inspect gasket for effective seal.
(b) Unlatch lid and remove from chest.
(c) Turn lid end-for-end (180° rotation in a horizontal plane), assemble with the chest and inspect for effective seal.
(d) Repeat (a) through (c), inclusive for a total of 2 cycles for first produced testing; or a total of 1 cycle for quality conformance testing or other testing as specified herein.
(e) With the lid lugs engaged by the hold-down devices and the latches disengaged, raise the lid to a 45 degree angle and allow it to fall to the chest. This test shall be repeated for a total of five times. Inspect the lid for structural failure and loosened gasket. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.2 Proof load. The chest, loaded with sand to an evenly distributed gross weight of 2,600 plus or minus 10 pounds and with the lid latched in place, shall be lifted free of all other support by the four lifting hooks and suspended for a period of not less than 10 minutes, then lowered to the ground. During and after the test, the lifting hooks, chest and lid shall be inspected for damage, permanent deformation, or failure, and the latches shall be observed for malfunction. The lid shall then be tested for one cycle as specified in 4.5.2.1. Nonconformance to 3.6.1 and 3.6.2 shall constitute failure of this test.

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4.5.2.3 Bar lift. With the chest loaded with sand to an evenly distributed gross weight of 1,200 plus or minus 5 pounds and the lid latched in place, an upward pull of 300 plus or minus 3 pounds shall be exerted on either bar. During and after the test, the bar, chest, and lid shall be inspected for damage, permanent deformation, or failure, and the latches shall be inspected for malfunction. The lid shall then be tested for one cycle as specified in 4.5.2.1. Nonconformance to 3.6.1 and 3.6.3 shall constitute failure of this test.

4.5.2.4 Drop. The chest, loaded with sand to an evenly distributed gross weight of 1,200 plus or minus 5 pounds minimum and with the lid latched in place, shall be suspended by the lifting hooks, with the bottom surface of the skids a minimum of 2 feet above a concrete floor or equivalent. The chest shall be released by means of a triggering device and shall fall free to the concrete surface. After impact, the lifting hooks, chest, and lid shall be inspected for permanent damage, deformation, or failure and the latches shall be examined for malfunction. The lid operation shall then be tested for one cycle as specified in 4.5.2.1, and the chest shall be subjected to tests specified in 4.5.2.3 and 4.5.2.4. Nonconformance to 3.6.1 and 3.6.4 shall constitute failure of this test.

4.5.2.5 Edgewise-drop (rotational). The chest, with the weight and lid conditions specified in 4.5.2.4, shall be tested as specified in FED. TEST METHOD STD. No. 101, Method 5008, except that the bottom surface of the skids shall be no less than 6 feet plus or minus 2 inches above the concrete surface. After impact, the chest and lid shall be examined for evidence of permanent deformation or failure, and the latches shall be examined for malfunction. The lid shall then be tested for one cycle as specified in 4.5.2.1. Nonconformance to 3.6.1 and 3.6.5 shall constitute failure of this test.

4.5.2.6 Towing. The chest, with weight and lid conditions specified in 4.5.2.4, shall be towed from either end (selected at random) by means of the towing eyes for a minimum distance of 400 feet across a concrete pavement at a speed of 5 to 7 miles per hour. During towing, the latches shall be observed for malfunction. After towing, the chest shall be subjected to the following examinations and tests:

(a) Visual examinations for unrepairable damage.
(b) Lid operation as specified for one cycle in 4.5.2.1.
(c) Lifting as specified in 4.5.2.3 and 4.5.2.4.

Nonconformance to 3.6.1 and 3.6.6 shall constitute failure of this test.

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4.5.2.7 Simulated rainfall. As a final test, the chest, with weight and lid conditions specified in 4.5.2.4, shall be tested as specified in FED. TEST METHOD STD. No. 101, Method 5009 for leakage of water using the simulated rainfall technique. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6 Inspection of packaging. Each chest shall be examined to determine compliance with the Section 5 requirements. Any defect shall be cause for rejection.

5. PACKAGING

5.1 Preservation and packing (see 6.4). Each chest, closed as provided for, shall be additionally secured with a strap applied around the chest through the forklift opening. The band shall be of nonmetallic strapping and a minimum width of 3/4 inch. Further preservation and packing shall be as required to assure carrier acceptance and safe delivery to destination at lowest ratings of the carrier.

5.2 Marking (see 6.4). Each chest shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The chest is intended for use as a storage, handling and shipping container for collapsible fuel tanks, ground cloths, fittings and repair kit.

6.2 Ordering data: Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced chest and number of chests required (see 3.2).

6.3 First produced chest. Any changes or deviations of production chests from the approved first produced chest during production will be subject to the approval of the contracting officer. Approval of the first produced chest will not relieve the contractor of his obligation to furnish chests conforming to this specification.

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6.4 Preservation and packing. Degrees of preservation and packing are not appropriate for this item. For purposes of preservation/packing level marking, the marking shall be as required by the customer.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 5430-0131

Review activity:
DLA - CS
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-52978 Chest: Collapsible Tank

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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